

HAT-P-37b

R-2517

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_180602 · created 2026-08-25T00:49:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458271.7469 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.046
Transit depth (Rp/Rs)^2	2.8 +/- 1.56 [%]
Orbital Inclination (inc)	83.88 +/- 2.48
Airmass coefficient 1 (a1)	1.19 +/- 0.067
Airmass coefficient 2 (a2)	-0.092 +/- 0.026
Scatter in the residuals of the lightcurve fit is	5.65 %
Best Comparison Star	#3 - [541, 228]
Optimal Aperture	3.22
Optimal Annulus	26.91
Transit Duration (day)	0.035 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.167 ± 0.046 vs 0.1394 (deviation 0.56σ)
Duration fit vs published	0.035 d vs 0.0983 d (deviation 2.36σ)
Mid-time O-C	11.1 min
Depth SNR	3.4
Residual scatter	5.65 %

Light curve

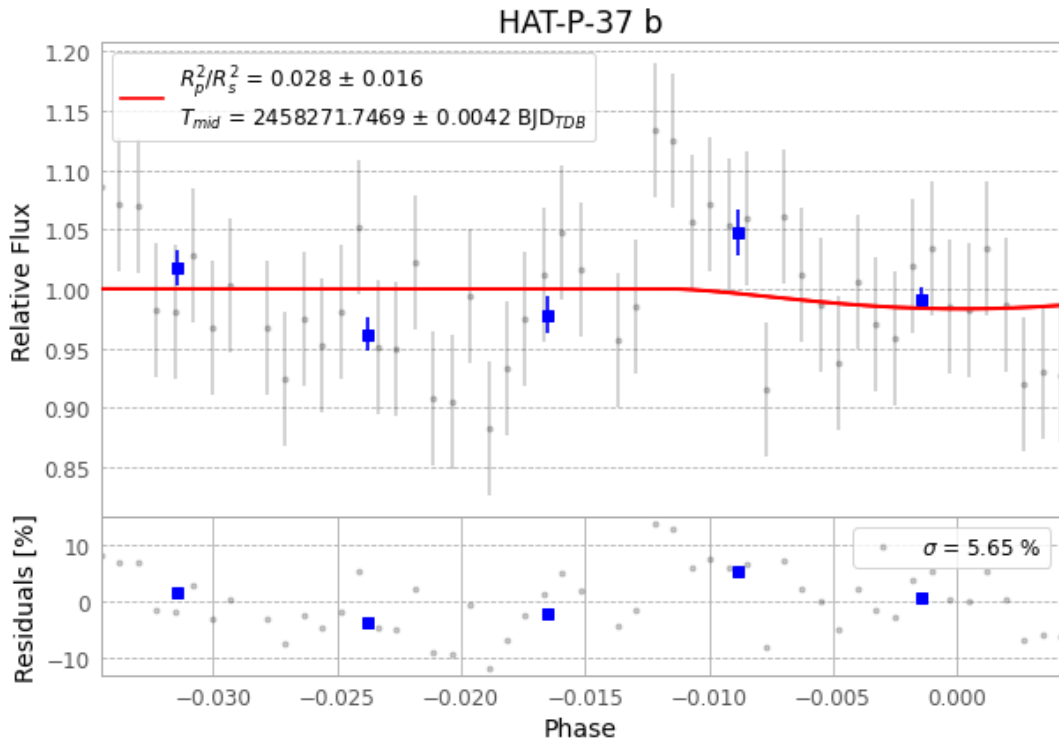


Figure 1 — FinalLightCurve_HAT-P-37 b_2018-06-01.png (SHA-256 ff1142b0d28ad95f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 984557bb675aa204...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), HATP-37180602033312.FITS → HATP-37180602060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-180602.log (29ce3a835f5b...), FinalLightCurve_HAT-P-37 b_2018-06-01.png (ff1142b0d28a...), FinalLightCurve_HAT-P-37 b_2018-06-01.csv (e4a5ea9cfbcc...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:49Z.